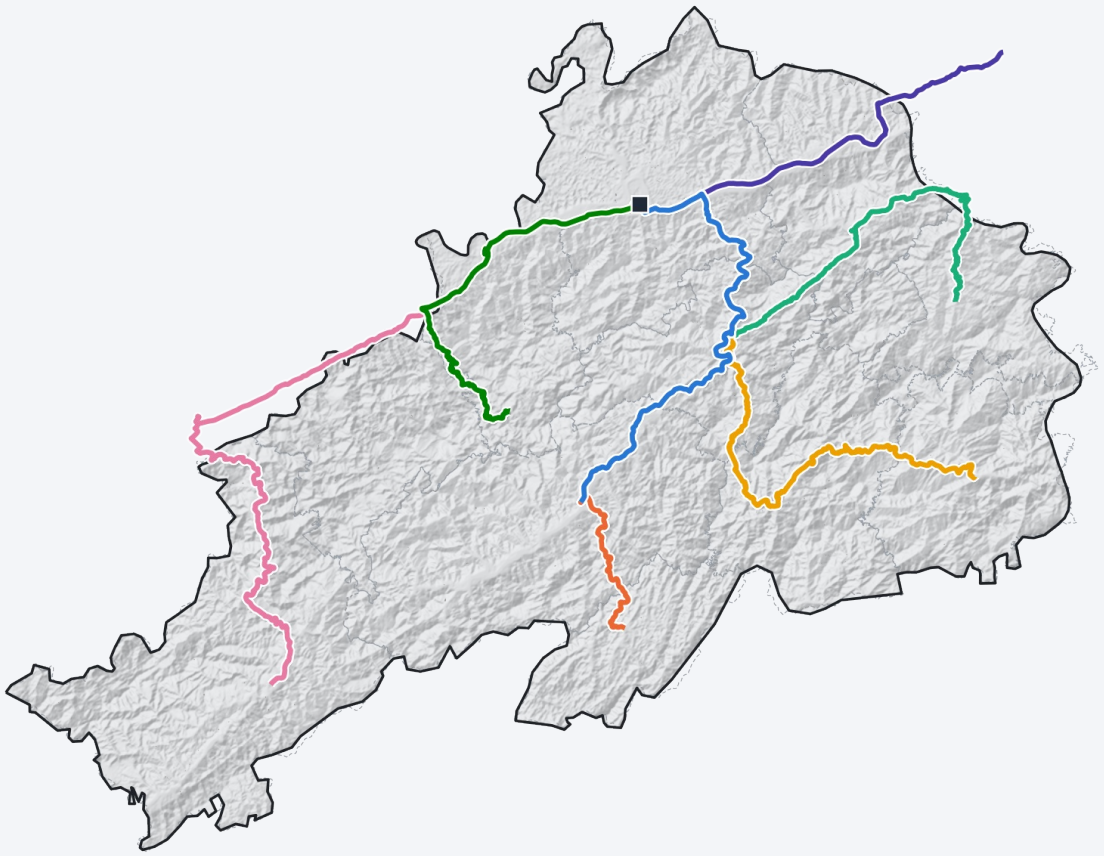
WORK IN PROGRESS · RESEARCH DESIGN

Research on Urban–Rural Public Transport in Xinhuang County

Run-level demand, overlapping calendar rhythms, and forecast-informed service planning in a mountainous county of western Hunan, China

Lilac
National University of Singapore



Seven urban–rural bus routes radiating from the county bus station (■), on terrain from Copernicus GLO-30 DEM

From a local service problem to a testable research design

I

Study area and data

- Xinhuang's seven radial bus routes
- Terrain, periodic markets and access
- Data in hand, data structure, public calendars

II

What prior research tells us

- Six themes: consensus, methods, limitations
- From prior practice to design choices
- Five gaps → four research questions

III

Framework and next steps

- Five-step research logic and how it evolved
- Surveys, calendar factors, run-level model
- Forecasting, service planning, timeline

Guiding question When rural calendar rhythms (school weeks, work weeks, holidays, Spring Festival) overlap, how much demand does each add to a single bus run — and how can forecasts of that demand make a fixed rural timetable more responsive?

Status: design stage. Every model result shown is a SYNTHETIC test of whether the method recovers known effects; no Xinhuang ridership results are reported yet.

I

Study area and data

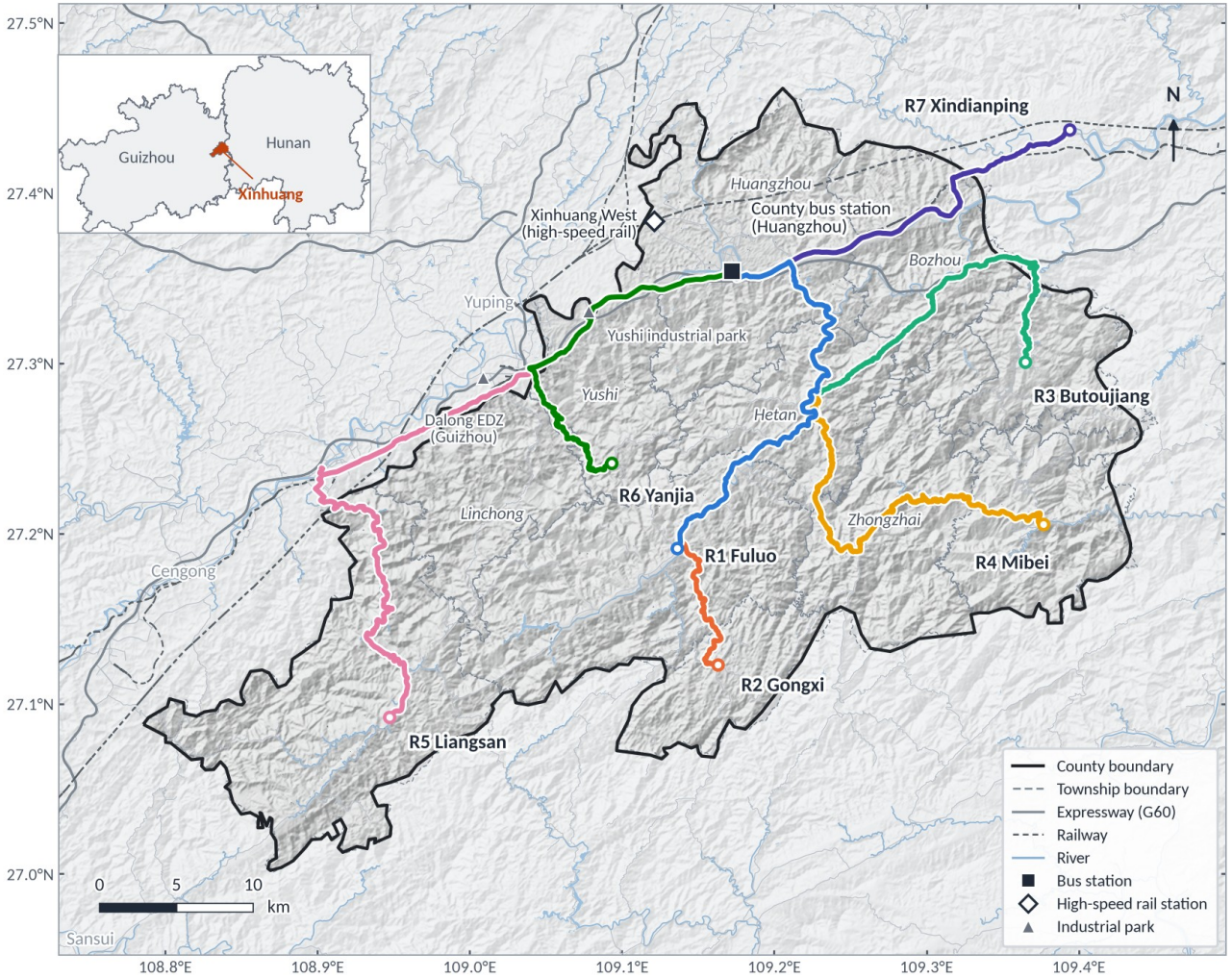
Where the problem sits: a mountainous border county, seven radial routes, and the data assembled so far.

I Study area and data

II What prior research tells us

III Framework and next steps

Xinhuang: a mountainous border county served by seven radial routes



Sources: OpenStreetMap contributors (ODbL); Copernicus GLO-30 DEM; 2024 Hunan village boundaries (townships dissolved). Routes routed on the OSM road graph.

≈1,500 km²

Xinhuang Dong Autonomous County, western Hunan, bordering Guizhou

11 + 1

townships plus one forest farm; 145 administrative villages

≈244,700

residents (WorldPop 2020 model estimate — not a census count)

7 routes

all radiating from the county bus station in Huangzhou, 26–61 km long

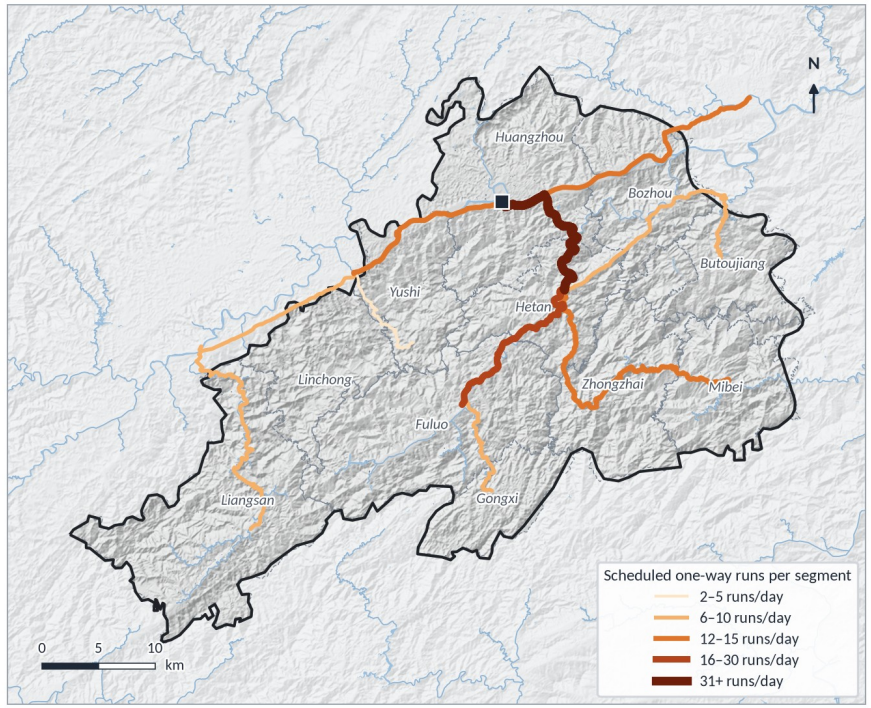
136 runs/day

one-way scheduled runs; hail-and-ride, no stop-level counts

A fixed timetable that thins out quickly beyond the trunk corridor

Route	Terminus	km	Drive min	Runs/day*	Outbound first–last	Inbound first–last
R1	Fuluo	33	45	13	7:00–17:00	7:00–17:00
R2	Gongxi	45	58	10	6:50–16:30	7:00–16:00
R3	Butoujiang	46	74	6	8:00–16:00	**6:30–14:30**
R4	Mibei	50	79	12	7:00–16:00	**6:10–15:00**
R5	Liangsan	61	95	10	7:10–16:20	**6:30–15:00**
R6	Yanjia	27	47	2	10:30–15:30	**6:50–12:30**
R7	Xindianping	26	34	15	6:40–16:30	7:00–17:00

* one-way runs per day in each direction. km and drive time from Gaode Maps; timetable from the operator's route document.

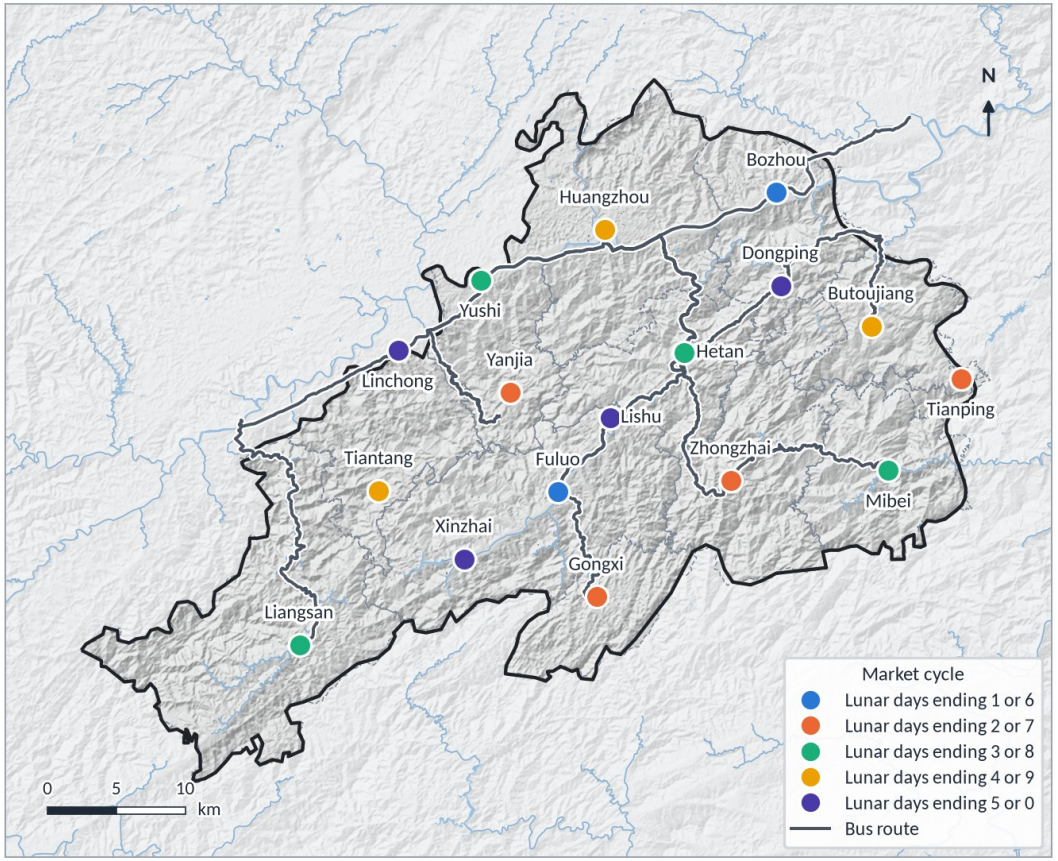


Scheduled one-way runs per road segment. This is supply, not demand: who fills these runs, and when, is what the ledger and surveys must reveal.

WHAT THE TIMETABLE ALREADY SHOWS (TO BE VERIFIED IN SURVEYS)

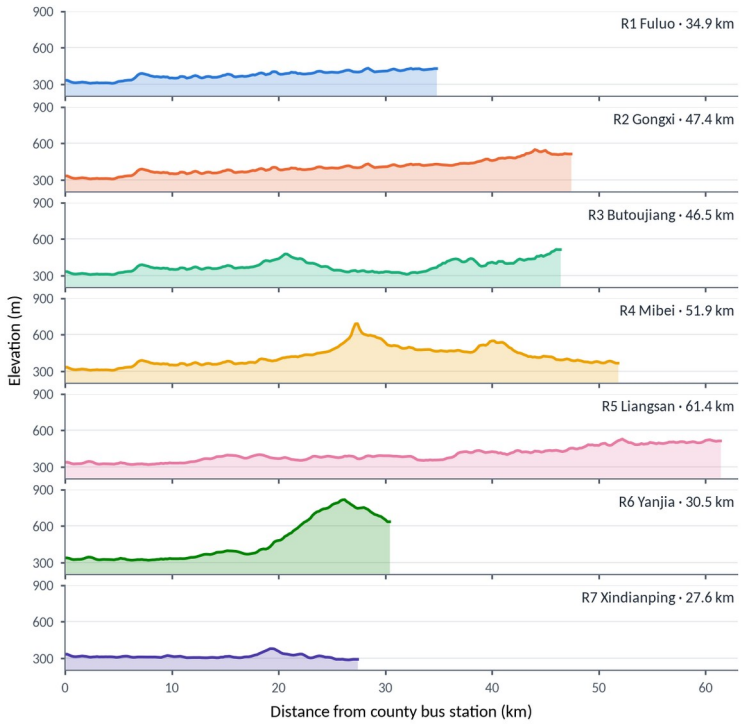
- **Supply is concentrated:** the 17-km Huangzhou–Hetan trunk carries 41 runs/day (R1–R4 overlap); branch ends see 2–10.
- **Early last buses into town:** on 4 of 7 routes the last inbound run leaves by 15:00 (R6 by 12:30), so an afternoon trip to the county town is impossible.
- **Same-day round trips are constrained** by outbound last runs at 15:30–17:00.
- **Data grain:** the ledger records total boardings per one-way run — no stop-level boardings, no origin–destination.

Terrain and periodic markets shape when and how villagers travel



17 MARKETS ON A 5-DAY LUNAR CYCLE

Neighbouring markets are staggered, so on every lunar day some route has a market within 2 km (R2: all ten day-digits; R7: four). Market days therefore enter the model as a route-specific baseline control, not as a core driver.



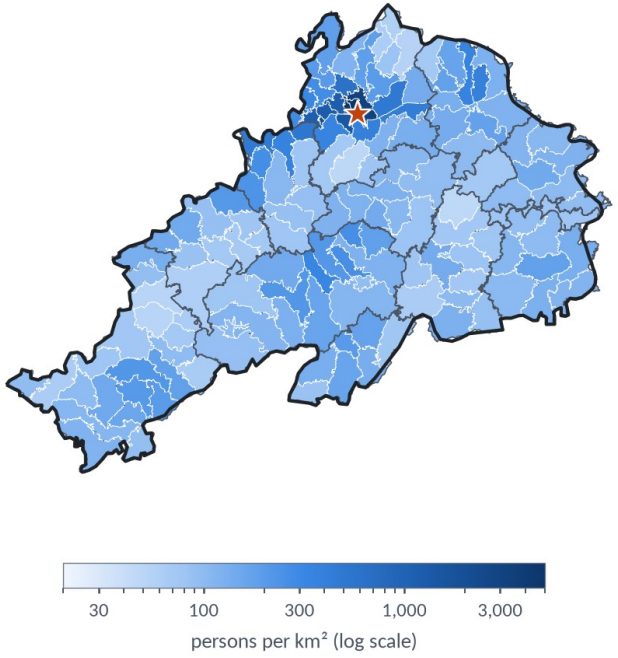
Elevation along each route (Copernicus GLO-30 DEM, 200 m sampling)

Route	Climb m	Curvy %	km/h
R1	419	1	44.0
R2	613	1	46.6
R3	672	**16**	37.3
R4	762	**18**	38.0
R5	667	7	38.5
R6	576	**28**	34.5
R7	191	0	45.9

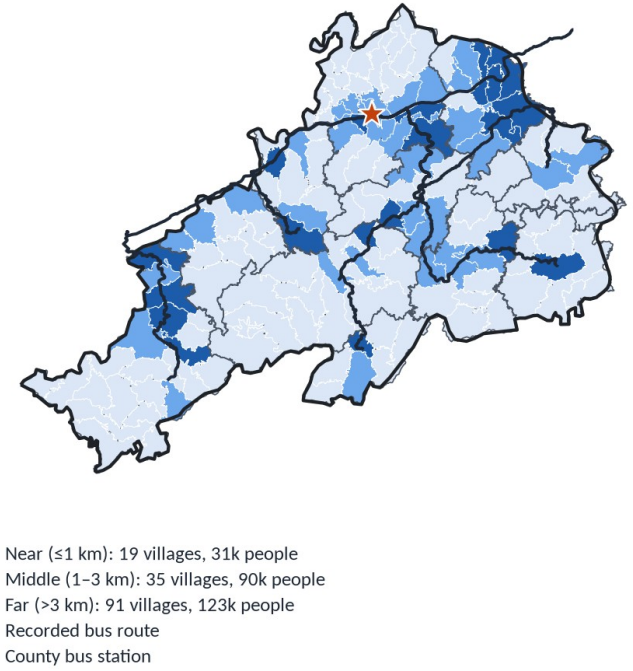
Cumulative climb, share of km in tight-curve sections, and navigation speed (v4 route analysis). Curvy, climbing routes are where on-board fieldwork observed motion sickness — now recorded per run (item Z4).

Half of the modelled population lives >3 km by road from a recorded route

a Modelled population density (WorldPop 2020)



b Network distance to a recorded route



Villages are the 2024 Hunan administrative-village polygons (145). Population: WorldPop 2020, 1 km, UN-adjusted, summed per village.

13% — **Near (≤1 km)**
19 villages · 31k people

37% — **Middle (1–3 km)**
35 villages · 90k people

50% — **Far (>3 km)**
91 villages · 123k people

CAVEATS BUILT INTO THE DESIGN

- "Recorded routes" = alignments confirmed so far; the route list will be checked with the operator.
- Network distance includes a snap segment, so strata lean "far": 88% of villages keep their stratum without it, 69% with straight-line distance.
- Sampling stays unbiased (known inclusion probabilities); access models will densify the road graph and report all three definitions.

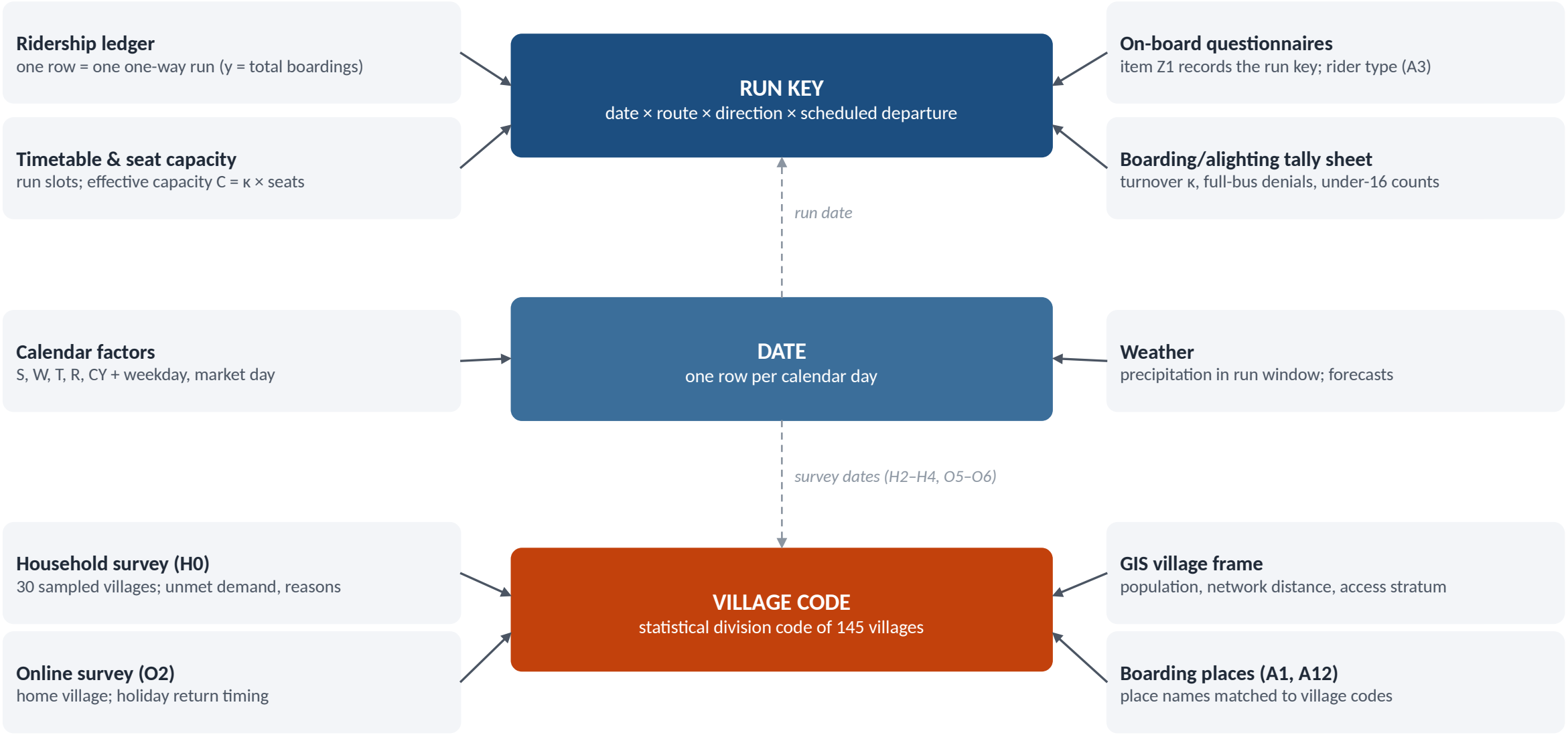
Implication: much rural demand never reaches the ledger. Unmet demand must be measured in villages, stratified by access.

Data assembled so far — and what each source can and cannot tell us

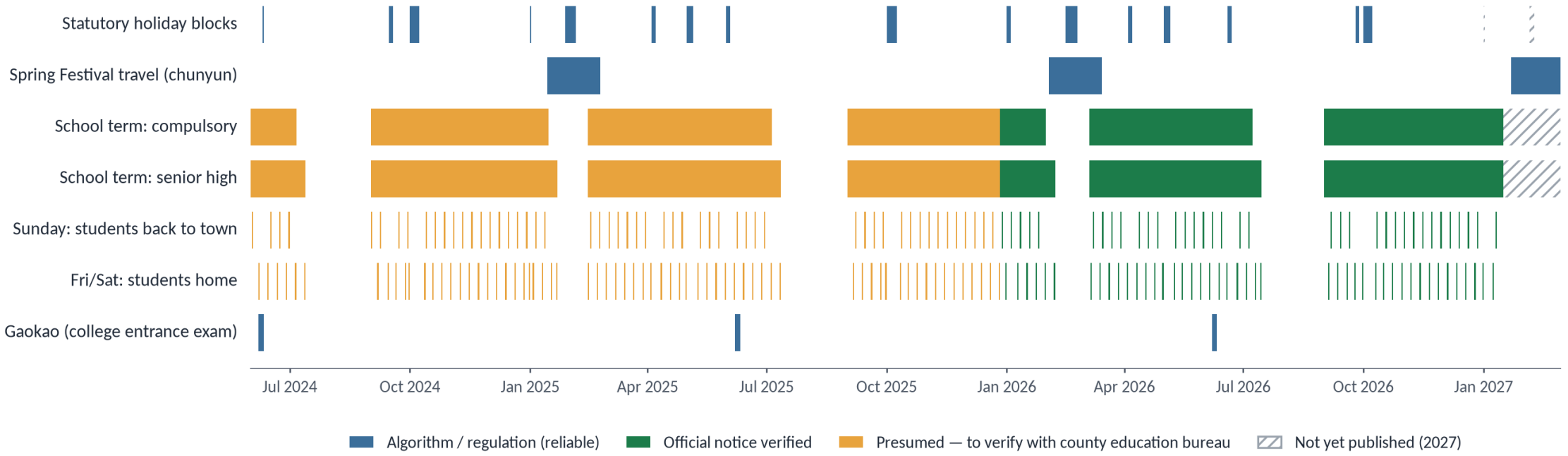
Source	Content and grain	Coverage / status	Role
Run-level ridership ledger (county bus station)	Total boardings per one-way run: date × route × direction × departure. No stop-level or OD data.	Exploration: 1 Jun 2024–30 Sep 2026 (852 days). Sealed test: Oct 2026–Feb 2027, released only after methods are frozen.	**Outcome for S2–S5**
Timetable and seat capacity	7 routes, 136 one-way runs/day → 115,872 run slots on the real calendar	Timetable in hand; per-run seat counts to be confirmed	**Sampling frame; censoring threshold**
Public calendar	Holidays & make-up workdays, lunar calendar, Spring Festival window, school terms, gaokao	Daily, Jun 2024–Feb 2027; each date graded by source reliability	**Five calendar factors + controls**
Weather	Precipitation and max. temperature in each run's time window; forecasts	Public reanalysis / station data	**Control; forecast input**
GIS	145 village polygons (2024), WorldPop 2020, OSM roads & facilities, GLO-30 DEM	Assessed file by file; townships rebuilt from villages	**Sampling frame, access, maps**
Summer on-board survey	Passengers aged ≥16 on sampled runs	Completed 15 Jul–31 Aug 2026	**RQ1; linked to runs**
Survey instruments (next waves)	4 questionnaires from a 47-item bank in 13 indicator groups	Designed (v5); pre-test pending	**RQ1, RQ4, diagnostics**
Literature base	250 English studies (incl. 25 method papers); 10 verified Chinese journal papers	Screened and depth-coded; CNKI search pending	**Part II**

Deliberately excluded: industrial-park shift rosters, hospital clinic schedules and macro market cycles — they cannot be obtained reliably day by day. If they matter, their effect falls into weekday or residual terms (a stated limitation).

Three keys link every source: run key, date, and village code



Most demand drivers can be known days in advance from public calendars



RELIABLE BY RULE

Statutory holidays and make-up workdays (State Council notices), lunar calendar, Spring Festival travel window, gaokao dates.

NEEDS VERIFICATION

School terms before December 2025 are presumed from provincial practice and must be confirmed with the county education bureau and the three county-town secondary schools.

NOT YET PUBLISHED

2027 winter-vacation and holiday dates. Winter survey strata are drawn now for planning and re-drawn with the same random seed once dates are announced.

II

What prior research tells us

Six themes. For each: what is established, which methods are used, what remains unresolved — and how this study responds.

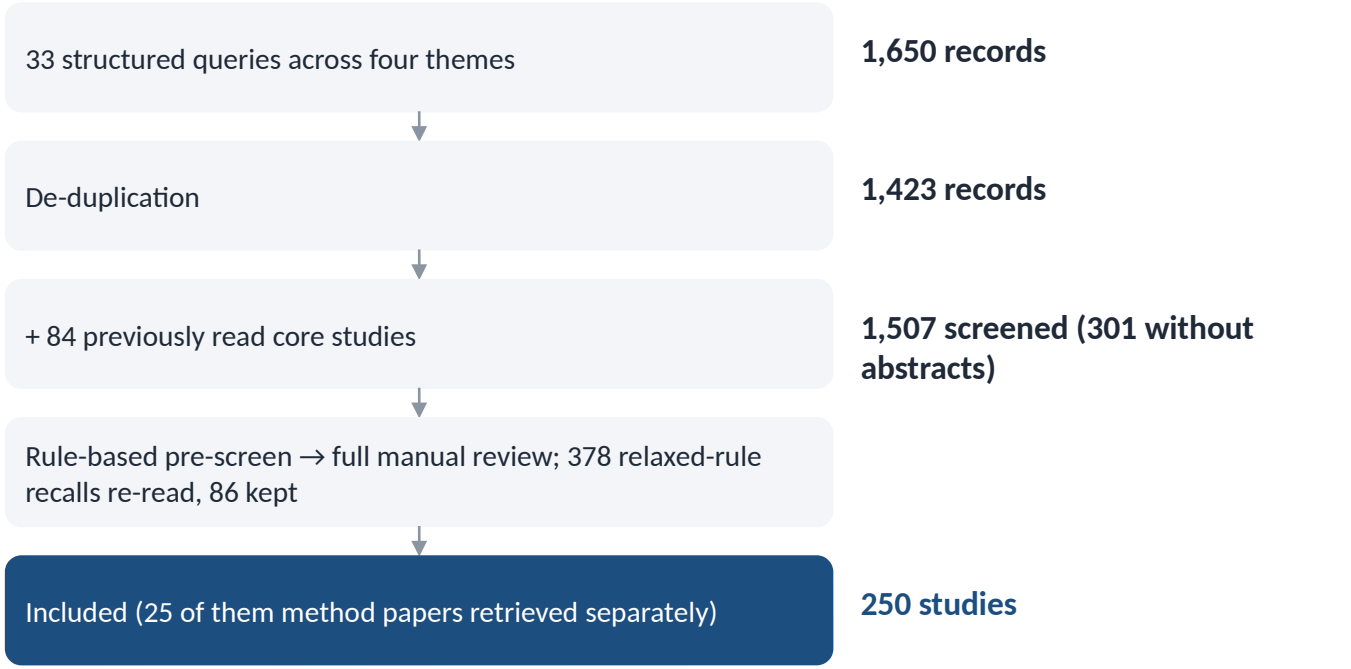
I Study area and data

II What prior research tells us

III Framework and next steps

How the evidence base was built

ENGLISH LITERATURE (OPENALEX, 26 SEP 2026)



- **By theme:** surveys & rider groups 48 · drivers 44 · forecasting 68 · service design 90; 60 explicitly rural or low-demand.
- **Rule vs. manual agreement was low** (Cohen's κ = 0.41, recall 0.46), so rules only narrowed reading; inclusion was decided manually.

CHINESE-LANGUAGE SUPPLEMENT

- **7 compliant retrieval routes** (no access controls bypassed): OpenAlex 16,448 + Crossref 8,800 records → 12,565 unique.
- **Concept tagging** (bus · rural · calendar time · method): 140 A-tier, 284 B-tier; 21 in core journals.
- **10 papers verified** against publisher pages (e.g., Progress in Geography, Human Geography, J. Highway & Transportation R&D).
- **Finding:** none studies run-level ridership of county urban–rural buses — within publicly reachable sources.
- **Pending:** seven CNKI search strings, run under an institutional account, before conclusions are final.

METHOD SOURCES

Traced to originals and DOI-checked: interaction scale, GA2M, boosting, ridge, demand unconstraining, GAMLSS, forecast reconciliation, chance constraints, SAA, integrated population models.

Who rides rural buses — and what surveys can and cannot see

WHAT IS ESTABLISHED

- Rural riders are concentrated among older adults, students, car-less and low-income households; trips serve health care, shopping, school and errands. [1–3]
- Survey mode decides who is visible: household surveys reach non-riders; on-board surveys tie answers to a specific run. [4]
- Non-response is selective (frequent travellers are hard to reach) but can be partly corrected by calibration weighting / raking. [5, 6]

METHODS IN USE

- On-board intercept, household and online surveys; mixed-mode designs need checks for mode effects. [4, 6]
- Rider typologies from smart-card regularities or latent-class analysis; satisfaction and service-quality scales. [7]

LIMITATIONS AND UNRESOLVED PROBLEMS

- Surveys are used to describe riders; their results rarely enter ridership models, and counts rarely calibrate surveys.
- Suppressed demand — trips not made for lack of a suitable bus — is invisible in boarding data. [8]
- Long attitude batteries overload older respondents on moving, winding buses (satisficing). [9]

→ HOW THIS STUDY RESPONDS

Surveys and ledger share a run key. On-board rider-type counts enter the joint model as a multinomial likelihood — used to diagnose the model's decomposition (idea borrowed from integrated population models [10]). Household surveys measure non-riders' unmet demand. Behaviour questions replace Likert batteries; on-board interviews are capped at 5 minutes.

KEY STUDIES BEHIND THE SYNTHESIS

[1] Pucher & Renne (2005) *Transportation*
Rural mobility and mode choice from the 2001 US NHTS: rural households are highly car-dependent; those without cars are disadvantaged.

[2] Berg & Ihlström (2019) *Social Sciences*
Public transport matters for everyday activities of rural residents without car access (Sweden).

[3] Yang & Cherry (2016); Espeland & Rowangould (2024) *TPT; JTG*
Rural riders skew older and lower-income; rural travel burdens and unmet need in the US.

[4] Stopher & Greaves (2007); Newmark (2014) *TR-A; JPT*
Household vs. transit-agency (on-board) survey practice and coverage.

[5] Kuhnimhof, Chlond & Zumkeller (2006) *TRR*
Non-response and selectivity bias in travel surveys.

[6] Deville & Särndal (1992); Kolenikov (2014) *JASA; Stata J.*
Calibration estimators; raking (iterative proportional fitting).

[7] Kieu, Bhaskar & Chung (2014) *QUT ePrints*
Passenger segmentation by travel regularity from smart-card data.

[8] Church, Frost & Sullivan (2000); Lucas (2012) *Transport Policy*
Transport and social exclusion: activities forgone for lack of transport.

[9] Krosnick (1991) *Appl. Cognitive Psychology*
Satisficing under cognitive burden degrades survey answers.

[10] Schaub & Abadi (2011) *J. Ornithology*
Integrated population models: counts and individual data share one likelihood.

What drives ridership: weather, seasons, holidays — and rural rhythms

WHAT IS ESTABLISHED

- Weather is the best-studied driver: rain and cold reduce ridership, non-linearly, varying by place, season and trip purpose; non-urban evidence is scarce. [1–5]
- Ridership carries layered seasonality — weekly, annual and holiday cycles superimposed. [6]
- Holidays and the Spring Festival reshape mobility strongly, with clear spatio-temporal structure — measured at inter-provincial, inter-city or metro scale. [7–9]

METHODS IN USE

- Holiday dummies in regression; group-mean comparisons; seasonal decomposition; day-of-week effects. [1, 6]

LIMITATIONS AND UNRESOLVED PROBLEMS

- One holiday dummy treats a 3-day Qingming break like the 8-day National Day break, and ignores direction (out before, back after).
- Weekly commuting is absorbed by day-of-week dummies: the model fits “Friday”, not “students going home”, and fails when school is out.
- Periodic markets are studied for trip satisfaction, not run-level ridership; weekly student and worker commuting is not studied as a driver. [10, 11]

→ HOW THIS STUDY RESPONDS

Encode drivers as explicit, directional calendar factors built from public sources: boarding-student weekly commute (S), worker weekly commute (W), term start/end surges (T), holiday return trips graded by cultural importance × length (R) and Spring Festival travel (CY). Weather, weekday and market day stay as baseline controls.

KEY STUDIES BEHIND THE SYNTHESIS

[1] Guo, Wilson & Rahbee (2007) *TRR*
Weather and Chicago bus/rail ridership; weekends and off-peak react more.

[2] Singhal, Kamga & Yazici (2014) *TR-A*
Hourly subway ridership in New York: weather effects differ by time of day and day type.

[3] Arana, Cabezudo & Peñalba (2014) *TR-A*
Smart-card study: rain and wind cut trips, especially leisure trips.

[4] Wei (2021); Wei & Liu (2022) *JTG; TBS*
Non-linear, threshold-like rainfall effects (“how wet is too wet?”).

[5] Stover & McCormack (2012) *JPT*
One of few non-urban studies: weather and bus ridership in Pierce County, WA.

[6] Kashfi, Bunker & Yiğitcanlar (2015, 2016) *JTG*
Complex seasonality plus weather in suburban daily ridership.

[7] Liu, Shi & Chen (2020); Huang, Liu & Zhao (2019) *IJGI*
Spring Festival changes metro, bus and taxi use in Shenzhen.

[8] Li et al. (2016); Wang et al. (2014); Zhou et al. (2020) *Sustainability; arXiv; IJGI*
National Spring Festival migration traced with big data.

[9] Zhao Z. et al. (2017, 2022) *Progr. Geography; Human Geography (in Chinese)*
Inter-provincial Spring Festival flows; festival inter-city travel.

[10] Hong, Liang & Zhu (2024) *Front. Public Health*
Rural periodic-market trips: satisfaction and road environment.

[11] Schönfelder & Axhausen (2010) *Ashgate*
Stable weekly and annual travel rhythms that differ between groups.

Days and stations dominate the evidence; decisions are made per run

WHAT IS ESTABLISHED

- Most ridership studies analyse network or line daily totals, station counts or cross-section loads. [1–4]
- Short-term forecasting targets urban metro or bus stations at minute-to-hour resolution with LSTM-type models, in China as elsewhere. [1–5]
- Station or section counts cannot simply be converted to whole-run totals; smart cards enable urban OD and segmentation. [6, 7]
- Rural demand estimates are mostly annual or cross-sectional, driven by population and land use. [8]

METHODS IN USE

- Count regression (Poisson, negative binomial) for small, over-dispersed counts; demand unconstraining when capacity censors observations. [9, 10]

LIMITATIONS AND UNRESOLVED PROBLEMS

- Daily totals hide within-day contrasts: a full Friday-afternoon run and an empty morning run average out.
- Supplementary runs are decided per run, yet evidence comes per day or per stop.
- Moving to runs brings small counts and censoring: a full bus records its capacity, not its demand.

→ HOW THIS STUDY RESPONDS

Unit = total boardings on one one-way run. Negative binomial counts with a censored likelihood (full bus ⇒ demand ≥ seats × turnover κ); daily totals are sums of runs. Real calendar evidence: boarding-student and worker commutes share 189 days but zero runs, so run-level VIFs are 1.28 versus 3.7 and 5.9 for daily totals.

KEY STUDIES BEHIND THE SYNTHESIS

[1] Zhang, Chen & Shen (2019) *IEEE Access*
Cluster-based LSTM for short-term metro passenger flow.

[2] Han, Wang & Ren (2019) *IJGI*
Hybrid optimised LSTM for short-term bus passenger flow.

[3] Liyanage, Abduljabbar & Dia (2022) *J. Urban Management*
Neural networks for bus demand from smart-card data.

[4] Alexandre, Bernardini & Viterbo (2023) *TRR*
Systematic review of machine learning for bus transport.

[5] Xie et al. (2018); Li G. et al. (2019) *in Chinese*
IC-card and LSTM station-level forecasting in Chinese cities.

[6] Ma, Ai & Zhang (2018) *J. Highway & Transp. R&D (in Chinese)*
Estimating bus cross-section flows under incomplete information.

[7] Egu & Bonnel (2020) *TR-A*
OD matrices from fare data vs. surveys in Lyon.

[8] Painter et al. (2007); TRB (2013) *TRR; TCRP Report 161*
Rural demand forecasting from population and land use.

[9] Jang (2005) *J. Transportation Engineering*
Count-data models for trip generation.

[10] Weatherford & Pölt (2002) *J. Revenue & Pricing Mgmt*
Unconstraining censored demand improves forecasts and revenue.

When peaks coincide: interaction, scale and identifiability

WHAT IS ESTABLISHED

- Calendar effects are usually estimated one factor at a time; co-occurring factors are treated as controls. Whether two peaks add, amplify or offset is rarely an estimand in transport research.
- Whether an interaction exists depends on the scale: no interaction on the multiplicative (log) scale implies interaction on the additive scale, and vice versa. [1]

METHODS IN USE

- Log-linear count models with interaction terms: coefficients live on the ratio scale. [2]
- Additive (passenger-count) models: directly operational, but need non-negativity and are harder to fit.
- Gradient boosting + SHAP; explainable boosting (GA2M) with pairwise shape functions. [3–5]
- Shrinkage (ridge) and distributional regression (GAMLSS) for many or unstable parameters. [6, 7]

LIMITATIONS AND UNRESOLVED PROBLEMS

- A main-effects log model assumes effects multiply, which itself creates additive-scale interaction.
- SHAP splits effects arbitrarily among correlated factors and gives no intervals; GA2M ignores censoring and collinearity.
- No approach first checks whether the data can identify an interaction at all.

→ HOW THIS STUDY RESPONDS

Define effects by counterfactual switches in passengers per run: independent effect, combined effect, and overlap = combined − Σ independent. Screen identifiability first (≥200 shared runs and ≥200 solo runs per factor). Choose among 12 candidate estimators by synthetic backtesting against known truth, not by preference.

KEY STUDIES BEHIND THE SYNTHESIS

[1] VanderWeele & Knol (2014) *Epidemiologic Methods*
Tutorial on interaction: additive vs. multiplicative scales.

[2] Jang (2005) *J. Transportation Engineering*
Poisson / negative binomial models for trip counts.

[3] Friedman (2001) *Annals of Statistics*
Gradient boosting machine.

[4] Ke et al. (2017) *NeurIPS*
LightGBM: efficient gradient-boosted trees.

[5] Lou, Caruana, Gehrke & Hooker (2013) *KDD*
GA2M: intelligible models with pairwise interactions.

[6] Hoerl & Kennard (1970) *Technometrics*
Ridge regression for non-orthogonal problems.

[7] Rigby & Stasinopoulos (2005) *JRSS-C*
GAMLSS: covariates for location, scale and shape.

[8] Weatherford & Pölt (2002) *J. Revenue & Pricing Mgmt*
Unconstraining capacity-censored demand.

Forecasting for decisions: distributions, calibration and fair evaluation

WHAT IS ESTABLISHED

- Decisions need predictive distributions: maximise sharpness subject to calibration, and evaluate with strictly proper scores (CRPS, log score); randomised PIT for counts. [1–3]
- Complex models do not reliably beat simple baselines; evaluation must roll forward in time to avoid leakage. [4, 5]

METHODS IN USE

- Intermittent-demand methods (Croston-type) for sparse series. [6]
- Conformal intervals around any point model, including adaptive versions for drifting series. [7, 8]
- Deep probabilistic models for many related series; hierarchical reconciliation for coherent totals. [9, 10]

LIMITATIONS AND UNRESOLVED PROBLEMS

- Transit forecasting is dominated by urban, short-term point forecasts; rural run-level distributions are rare.
- Run and daily forecasts are built separately and do not add up; daily intervals are too narrow because runs share county-wide shocks.
- Deep models need far more data than a county ledger provides.

→ HOW THIS STUDY RESPONDS

One run-level model with a county-wide day shock: daily totals are sums of run-level forecast samples, so their intervals carry the correlation between runs. Evaluate CRPS and 90% coverage overall and by route, time and event against seasonal-naïve and pipeline baselines — on rolling origins and one sealed winter.

KEY STUDIES BEHIND THE SYNTHESIS

[1] Gneiting, Balabdaoui & Raftery (2007) *JRSS-B*
Probabilistic forecasts: calibration and sharpness.

[2] Gneiting & Raftery (2007) *JASA*
Strictly proper scoring rules (CRPS, log score).

[3] Czado, Gneiting & Held (2009) *Biometrics*
Predictive model assessment for count data (PIT).

[4] Makridakis, Spiliotis & Assimakopoulos (2019) *IJF*
M4 competition: complexity does not guarantee accuracy.

[5] Hewamalage, Ackermann & Bergmeir (2022) *DMKD*
Forecast-evaluation pitfalls and best practice.

[6] Croston (1972); Petropoulos et al. (2013) *JORS; IMDS*
Forecasting intermittent demand.

[7] Lei et al. (2017); Romano et al. (2019); Angelopoulos & Bates (2023) *JASA; NeurIPS; FnT ML*
Distribution-free conformal prediction.

[8] Gibbs & Candès (2021); Barber et al. (2023) *NeurIPS; Ann. Stat.*
Conformal inference under distribution shift.

[9] Salinas, Flunkert & Gasthaus (2019) *IJF*
DeepAR: autoregressive recurrent probabilistic forecasts.

[10] Wickramasuriya, Athanasopoulos & Hyndman (2019) *JASA*
Optimal reconciliation (MinT) needs the error covariance.

Fixed, flexible or hybrid service under uncertain demand

WHAT IS ESTABLISHED

- Classical planning sets frequencies and timetables to balance ridership and cost; newer work adds robust or stochastic demand. [1–4]
- Rural demand-responsive transport (DRT) is widely trialled, but many schemes fail on cost and low use; there is an efficiency–equity dilemma, and older users struggle with booking. [5–8]
- Research is moving toward hybrid fixed + flexible designs. [9, 10]

METHODS IN USE

- Frequency-setting and timetabling optimisation; chance-constrained programming solved by sample average approximation; simulation. [1, 11]

LIMITATIONS AND UNRESOLVED PROBLEMS

- Hybrid designs usually assume idealised or urban demand rather than actual probabilistic forecasts.
- Rural DRT evidence is mostly European; demography, roads and operating institutions differ from a Chinese mountain county run by a single bus station.
- Point forecasts fed into optimisation ignore risk on high-demand days.

→ HOW THIS STUDY RESPONDS

Keep a fixed core timetable (no booking needed). Add a day-ahead reserve sized from the daily-total forecast, and forecast-triggered supplementary runs when Pr(demand > effective capacity) > α. Report a cost–denied-boarding frontier instead of one “optimal” plan; triggers come from calendar information known days ahead, not from bookings.

KEY STUDIES BEHIND THE SYNTHESIS

[1] Newell (1971); Ceder (1984) *Transp. Science; TR-A*
Dispatching policies; frequency setting from passenger counts.

[2] Hadas & Shnaiderman (2012) *TR-B*
Minimum-cost frequencies with stochastic demand and travel time.

[3] Durán-Micco & Vansteenwegen (2021) *Public Transport*
Survey of network design and frequency setting.

[4] Gkiotsalitis et al. (2021); Xia et al. (2023) *TR-C*
Frequencies and timetables under demand uncertainty.

[5] Mageean & Nelson (2003); Laws et al. (2009); Vansteenwegen et al. (2022) *JTG; JPT; TR-C*
DRT evaluations and review.

[6] Currie & Fournier (2020) *Res. Transp. Economics*
Why most DRT / micro-transit schemes fail.

[7] Schasché, Sposato & Hampl (2022) *Transport Policy*
Rural DRT: conflicting expectations, weak acceptance.

[8] Jittrapirom et al. (2019); Knierim & Schlüter (2021) *RTBM; JTG*
Older and less-mobile people's attitudes to DRT.

[9] Nourbakhsh & Ouyang (2011); Zhao, Sun & Cats (2021); Calabrò et al. (2023) *TR-B; Transportmetrica A; TR-A*
Structured flexible and joint fixed + DRT designs.

[10] Sörensen et al. (2020) *Transport Policy*
How much flexibility does rural public transport need?

[11] Charnes & Cooper (1959); Kleywegt et al. (2002) *Mgmt Science; SIAM J. Optim.*
Chance constraints; sample average approximation.

From prior practice to this study's design choices

Prior practice	Why it falls short for Xinhuang	Design choice in this study
Daily or station totals	Opposite-direction flows on the same day blur together; decisions are per run	**Run-level unit: total boardings per one-way run**
One holiday dummy	3-day and 8-day holidays differ in return-trip volume and direction	**Holiday intensity by type × length (hierarchical type effect)**
Day-of-week dummies	Fit “Friday”, not “students going home”; break down in vacations	**Directional student (S) and worker (W) factors switched by the school calendar**
Log-scale interaction terms	Answer a ratio-scale question; operators need extra passengers	**Passenger-scale overlaps; scale chosen by likelihood**
All interactions / ML + SHAP	Unidentifiable pairs; arbitrary attribution; no intervals	**Identifiability screen (≥200 runs) before estimation**
Full buses read as demand	Peaks underestimated (−14% in synthetic tests)	**Censored likelihood; capacity = seats × turnover κ**
Pipeline: effects → point forecast → optimisation	Uncertainty lost between steps; daily intervals too narrow	**One joint model; decisions drawn from forecast samples**
Surveys as description only	Never checked against ridership; unmet demand unseen	**Composition likelihood as diagnostic; household survey for unmet demand**

Are these gaps real? A three-step check beyond keyword counts

1 Correct hit counts for precision

“Transit ridership forecasting”: ≈1,602 relevant studies (95% CI 1,220–1,919). “Rural transit × ridership forecasting”: ≈16 (11–23). “Rural transit × probabilistic forecasting”: 1 hit, not relevant.

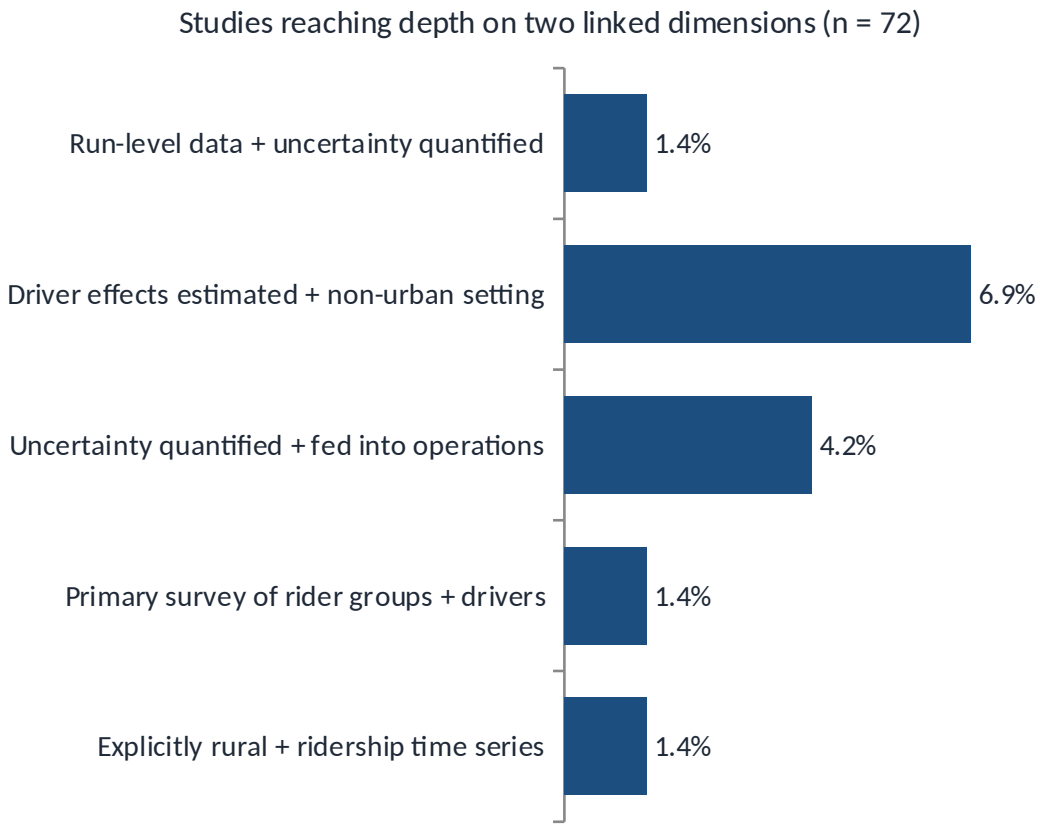
2 Read narrow searches in full

Holidays or Spring Festival × ridership: 6 relevant, all rail or urban. Rural DRT demand forecasting: 3, all cross-sectional. Market days × buses: 4, none on run-level ridership.

3 Code research depth, not mentions

72 stratified studies scored 0–2 on six dimensions (rider surveys, driver effects, time grain, uncertainty, link to operations, rural setting).

What can — and cannot — be concluded. Each theme is well developed, but deep links between adjacent steps are rare. This is not a claim that “no study exists”: the CNKI search is pending and depth coding had one coder (a second coder will check a sample).



95% Wilson intervals: 1.4% (0.2–7.5) · 6.9% (3.0–15.2) · 4.2% (1.4–11.5).

Five gaps lead to four research questions

Gap		RQ1	RQ2	RQ3	RQ4
G1	Unit stuck at day or station; no whole-run measure		●	●	
G2	Drivers dominated by weather; predictable rural calendar rhythms understudied	●	●		
G3	Single-factor estimates; independent vs. combined effects of overlapping factors not separated		●		
G4	Stages modelled separately; uncertainty not passed downstream			●	●
G5	Surveys disconnected from ridership records	●	●		

The questions chain: RQ1's rider groups explain RQ2's effects; RQ2's effects drive RQ3's forecasts; RQ3's distributions are RQ4's inputs.

RQ1 · S1

Which rider groups use the county's urban–rural buses, and where does the current timetable leave demand unmet — by group and by village?

RQ2 · S2–S3

When predictable rural calendar factors overlap, how large are their independent and combined effects on a single run — additive, amplifying or offsetting?

RQ3 · S4

Can run-level predictive distributions be well calibrated, and stay calibrated when summed to daily totals?

RQ4 · S5

With capacity-censored demand, how should forecast distributions set chance-constrained supplementary runs, trading cost against denied boardings?

III

Research framework and next steps

Five linked steps — from surveys to service planning — implemented as one run-level model, with methods chosen by backtest and frozen before testing.

I Study area and data

II What prior research tells us

III Framework and next steps

Five linked steps from surveys to service planning

1 Observe

S1 · RQ1

Who rides on which days? Do volumes and rider groups differ by day type in recurring ways? What problems do riders and non-riders report?

DATA · METHOD
On-board (summer, fall weekends, winter), household and online surveys; tally sheets

OUTPUT
Rider groups by day type; problem ranking; unmet demand; turnover κ

2 Explain

S2 · RQ2

Do market days, weather, public and lunar holidays, and school vacations change demand — and by how much?

DATA · METHOD
Public calendars, school terms, weather, timetable

OUTPUT
Daily factor calendar; identifiability table

3 Quantify

S3 · RQ2

Which factors move demand on an individual run, and how large are independent and combined effects?

DATA · METHOD
Run ledger + factors; censored count model, passenger scale

OUTPUT
Independent, combined and overlap effects with intervals

4 Forecast

S4 · RQ3

Can tomorrow's factors predict each run's demand 1–7 days ahead — with honest uncertainty?

DATA · METHOD
Fitted model + calendar and weather forecasts

OUTPUT
Run and daily-total predictive distributions

5 Plan

S5 · RQ4

How should service adapt: fixed core plus flexible runs scaled to the day's forecast and to survey-identified problems?

DATA · METHOD
Forecast samples, seats, costs, survey preferences

OUTPUT
Reserve and triggered-run rules; cost–denial frontier

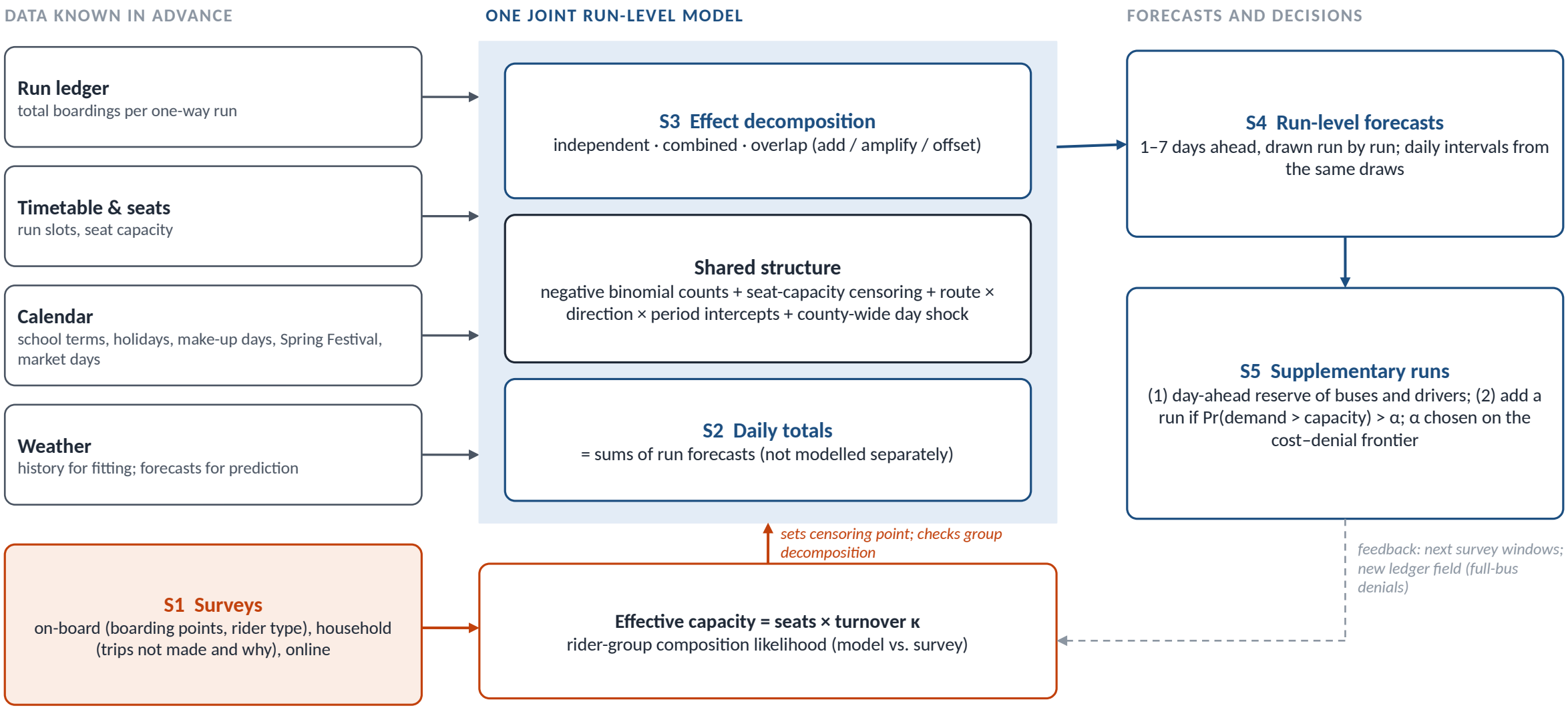
Feedback: supplementary-run results decide where the next survey wave samples more densely and which ledger fields are added (e.g., full-bus denials).

How the design evolved: six revisions and the evidence behind them

Earlier design	Current design	Why the change
Stages 2–5 as a sequential pipeline	**One run-level joint model; stages are its uses**	Pipelines drop uncertainty. Synthetic test: daily 90% intervals cover 0.39 (pipeline) vs 0.87 (joint)
Drivers incl. industrial-park shifts and clinic schedules	**Only reliably obtainable variables (21 kept, 3 dropped); worker weekly commute added**	Rosters cannot be obtained day by day; the commute of park and township staff can be encoded from calendars
Market day as a core driver	**Market day as a baseline control**	Advisory review; the survey's shopping/market purpose checks whether a control is enough
“Which factors matter?”	**“How large are independent and combined effects when factors overlap?”**	The real timetable × calendar shows heavy overlap by day but separation by run and direction
Worker commute coded in the students' direction	**Workers leave town Sunday p.m., return Friday p.m. / Saturday a.m.**	Correction raised student-only runs from 33 to 5,104; the earlier “inseparable” conclusion was withdrawn
Surveys as the main source for effects	**Surveys for description, diagnosis and external constraints**	Synthetic test: adding 400–600 questionnaires left overlap-effect error unchanged (0.038 → 0.038); ≈5,500 would be needed to detect misspecification

Lesson from the direction error: a mis-coded factor still fits well — it just assigns effects to the wrong group. Encodings are therefore checked against survey-reported departure times and dispatcher experience before estimation.

One run-level model links effects, forecasts and decisions



Four surveys, each assigned a job the ledger cannot do

Survey	Sample design	Window	Planned n	Main job
On-board, summer 2026	Passengers ≥16 on sampled runs (legacy instrument)	15 Jul–31 Aug 2026 · done	collected	RQ1; linked to runs by bridging rules
On-board, fall-term weekends	6 strata: calendar factor × direction (A1–A6), 30 runs, systematic sampling	16 Oct–20 Dec 2026	240	Test student vs. worker encodings
On-board, winter & Spring Festival travel	5 strata (B1–B5), 129 runs; re-drawn when 2027 dates are published	1 Jan–28 Feb 2027	≈1,030	Composition check; holiday timing
Household (village)	3 access strata; PPS villages (30 + 6 reserve); 15 addresses per village; no substitution	With the fall wave	360 households	Non-riders; unmet demand; reasons
Online	Out-migrant workers, students' parents; channel quotas (non-probability)	Around holidays	per quota	Return timing; cross-checks
On-board, summer 2027	Redesigned instrument, factor strata	Jul–Aug 2027	per design	Robustness

TALLY SHEET ON EVERY SAMPLED RUN

Stop-by-stop boardings and alightings give turnover κ, full-bus denials, and counts of under-16 students who are not interviewed.

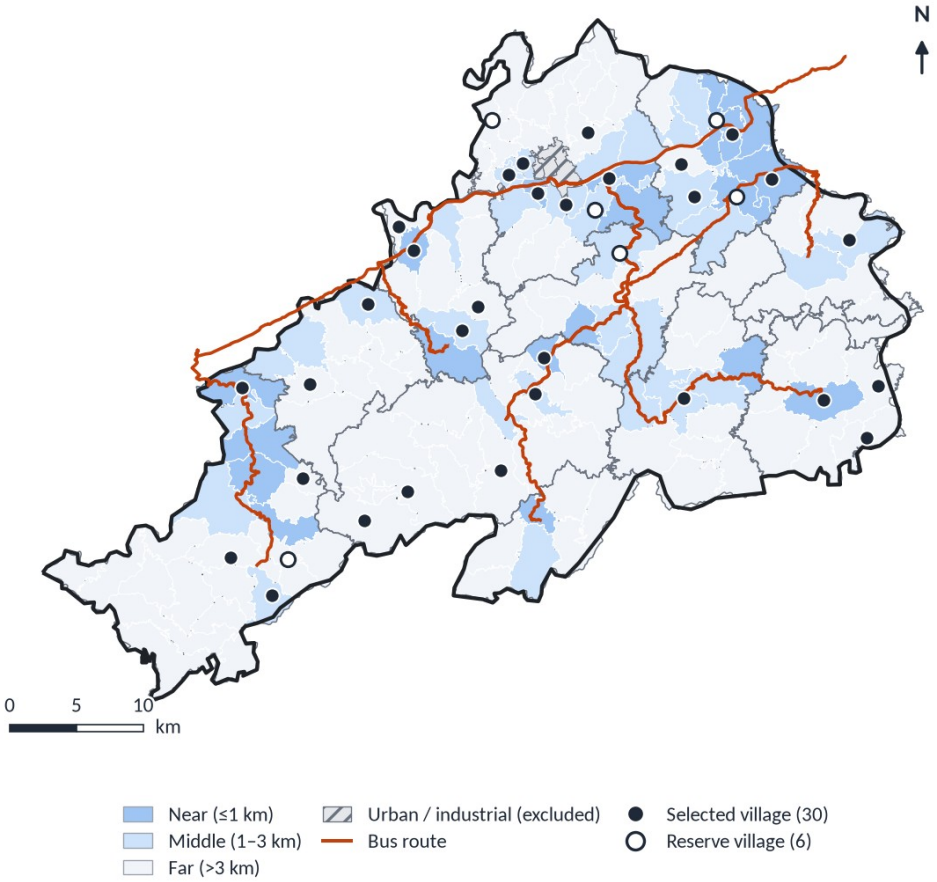
SHORT, BEHAVIOUR-BASED INSTRUMENTS

56 respondent items from a 47-item bank in 13 groups. Rule-based length: on-board 4.8 / 4.7 min (cap 5), household 5.7 (cap 15), online 3.5 (cap 10).

EVERY ITEM EARNS ITS PLACE

Each item must enter a stage and serve an RQ: of 53 earlier items, 33 kept or rewritten, 11 merged, 9 dropped; 10 added.

Identifying current problems — riders and non-riders, by village access



Villages drawn by systematic PPS within access strata (seed 20261001); one certainty village (Shawan).

SAME PROBLEM QUESTION IN ALL THREE SURVEYS

"Which one of these affects your bus use most?" — forced single choice replaces an 11-item Likert battery:

Too few runs	Last bus too early
First bus too late	Don't know when the bus comes
Bus full, could not board	Long, winding ride / motion sickness
Home too far from the boarding point	No obvious problem

UNMET DEMAND, MEASURED AS BEHAVIOUR

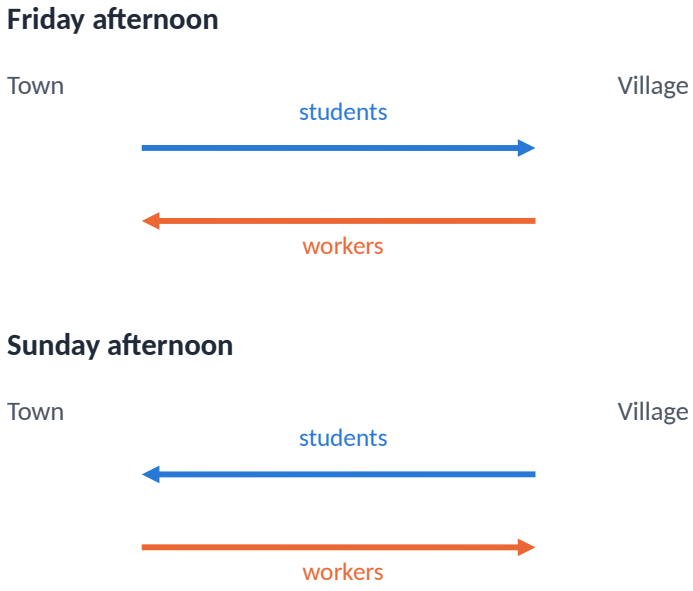
- **On-board:** trips given up or postponed in the past month; what you would do if this run did not operate.
- **Household:** household bus trips in the past 7 days; reasons for rarely riding; trips not made for lack of a bus.
- **Online:** arrived in the county town but no bus home; preferred departure times around holidays.

Pre-registered checks: boarding-student share on in-term Friday/Sunday runs exceeds vacation runs; far villages report higher unmet demand. County-wide 95% CI half-width ≈ 6.3 pp (p = 30%, ICC 0.05).

Turning holidays, school calendars, markets and weather into run-level factors

Driver in the original plan	Encoding on each run (date × time × direction)	Role
School term: boarding students' weekly trip	S — Sunday p.m. inbound (back to school); last school day p.m. outbound (home); zero in vacations	**Core factor**
Weekly commute of park and township workers	W — Sunday p.m. outbound; Friday p.m. inbound; first rest day a.m. inbound (weight 0.5); not tied to school terms	**Core factor**
Winter and summer vacations: start and end	T — eve of term inbound, first vacation day outbound (next day 0.4–0.5); vacations also switch S off	**Core factor**
Gregorian holidays (New Year, Qingming, Labour Day, National Day) and lunar holidays (Dragon Boat, Mid-Autumn)	R — out before, back after; intensity by holiday type × √(days / 7), estimated with a hierarchical type effect	**Core factor**
Spring Festival (lunar new year)	CY — days −20…0 and +2…+20; bell-shaped peaks 3 days before and around day 7	**Core factor**
Market days (lunar 5-day cycle)	Market-day indicator for towns along the route	Control
Weather	Precipitation in the run window, max. temperature, extreme-weather flag	Control
Weekday, make-up workdays, year	Day-of-week and adjusted-workday indicators; year terms	Control

OPPOSITE DIRECTIONS ON THE SAME DAYS



Same day, opposite buses: the two commuting groups share 189 days but never the same run — the key to separating their effects.

Real timetable × calendar: factors overlap by day but separate by run

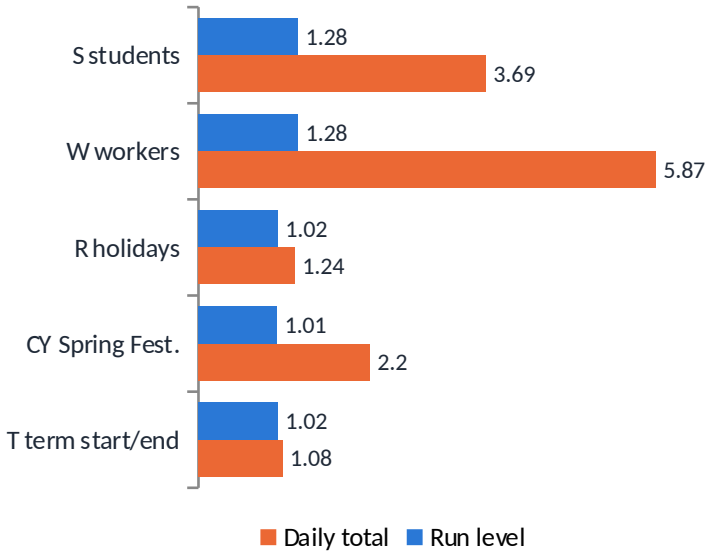
a Runs affected by both factors

	S	W	R	CY	T
S	6,210	0	845	229	32
W	0	12,286	69	437	298
R	845	69	3,536	0	0
CY	229	437	0	5,440	816
T	32	298	0	816	2,040

b Days on which both occur

	S	W	R	CY	T
S	191	189	26	12	6
W	189	354	38	30	16
R	26	38	51	0	0
CY	12	30	0	80	12
T	6	16	0	12	30

c Variance inflation (main-effects model)



- **Separated by direction.** Students and workers co-occur on 189 of 191 student-commute days but on 0 runs; VIF drops from 3.7 / 5.9 (daily) to 1.28 (run).
- **Some pairs never meet.** Holiday returns never coincide with Spring Festival travel or term surges: no interaction exists to estimate.
- **Support screen before estimation.** A pair gets an overlap term only with ≥200 shared runs and ≥200 solo runs each: 5 of 10 pairs pass (S×R, S×CY, W×CY, W×T, CY×T); the rest are reported as sums of main effects, flagged “low support”.

A censored count model on the passenger scale

$$Y_i \sim \mathrm{NegBin}(\mu_i, \theta), \quad y_i = \min(Y_i, C_i), \quad C_i = \kappa_r \times \mathrm{seats}_i$$

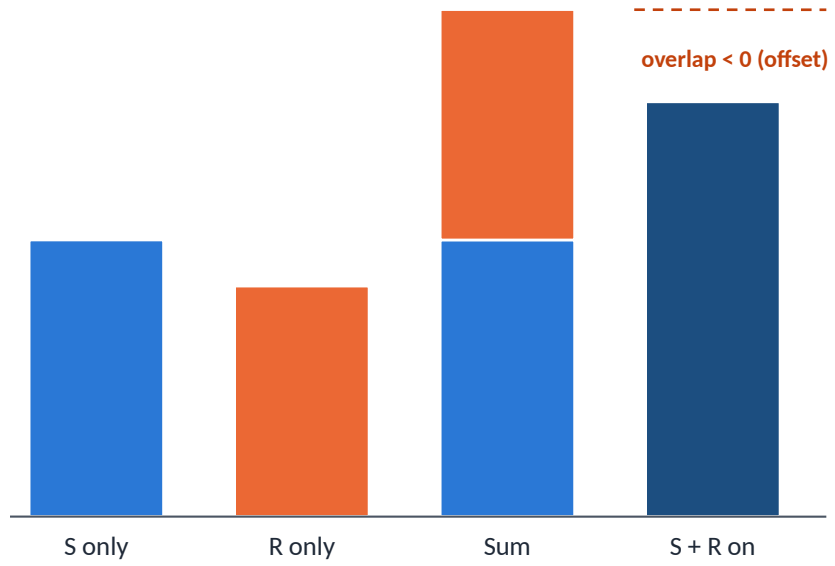
$$\mu_i = \exp(\alpha_{r\times dir\times period} + \gamma x_i + u_d) \times [1 + \Sigma\, \rho_g f_{gi} + \Sigma_{\Pi}\, \delta_{pq} f_{pi} f_{qi} + \Sigma\, \rho_{gt} f_{gi} 1_{year=t}]$$

- **Baseline:** hierarchical route × direction × period intercepts; controls x (weekday, market day, weather, year); county-wide day shock $u_d \sim N(0, \sigma^2)$.
- ρ_g: extra passengers, as a share of baseline, when group g's factor is fully on; δ_{pq}: overlap, only for screened pairs Π; ρ_{gt}: ridge-shrunk drift for 2025–26.
- **Full buses** contribute Pr(Y ≥ C) to the likelihood; turnover κ comes from tally sheets and is resampled in every draw.

Why the passenger scale: each term is "an extra group of people", maps one-to-one to survey rider types, and is what supplementary-run decisions need. Ratio-scale results are reported alongside.

EFFECTS AS COUNTERFACTUAL SWITCHES

(illustrative, passengers per run above baseline)



- **Independent:** switch on one factor only.
- **Combined:** switch on all factors in an overlap cell.
- **Overlap = combined − Σ independent:** >0 amplify; <0 offset (e.g., students ride once on a holiday Sunday); ≈0 additive.

SYNTHETIC DATA · tests the method, not Xinhuang ridership

Choosing the estimator by synthetic backtest, not by preference

Method	Combined-effect error				Independent-effect error			
	A	B	C	D	A	B	C	D
M1 Log · main effects	0.10	0.08	0.16	0.10	0.21	0.06	0.20	0.21
M2 Count · main effects	0.06	0.14	0.12	0.05	0.08	0.11	0.08	0.08
M3 Log · interactions	0.08	0.10	0.11	0.08	0.16	0.06	0.16	0.16
M4 Count · overlaps	0.06	0.11	0.10	0.06	0.08	0.09	0.07	0.07
M5 Log · inter. · censored	0.05	0.07	0.06	0.06	0.12	0.04	0.12	0.11
M6 Count · overl. · censored	0.05	0.09	0.09	0.06	0.03	0.13	0.03	0.04
M7 Gradient boosting	0.15	0.18	0.14	0.15	0.14	0.18	0.15	0.15
M8 GA2M (EBM)	0.15	0.18	0.15	0.16	0.11	0.12	0.11	0.12
M9 M6 + yearly drift	0.05	0.09	0.09	0.05	0.04	0.13	0.04	0.04
M10 M9 + screen (penalised)	0.05	0.09	0.09	0.05	0.04	0.13	0.04	0.04
M11/12 Screened, unpenalised	**0.07**	**0.09**	**0.06**	**0.07**	**0.04**	**0.05**	**0.04**	**0.04**

Relative RMSE on two-factor cells; 50 synthetic ledgers per world on the real timetable × calendar (115,872 runs). Worlds: A additive with absorption · B log-scale synergy · C sparse-cell synergy · D effects drift by year. M11/12 = identifiability screen, unpenalised overlaps, censoring, yearly drift, scale chosen by likelihood.

- **Forecast scores cannot pick the method:** CRPS differs by ≤1% across count models, yet effect estimates differ several-fold.
- **Right scale beats more interactions:** independent-effect error 0.034 (count scale) vs 0.117 (log scale) in world A.
- **Censoring removes bias:** Spring Festival cell −14.2% → −0.8%.
- **Penalising overlaps hides real absorption:** combined-effect bias +7.1% vs −0.5% unpenalised.

Frozen choice (1 Jan 2027): M11/12
Best worst case on both criteria (≤0.09 combined, ≤0.05 independent). Intervals widened ×2.3 after backtest; leave-one-world-out coverage ≥0.93. M10, M6 and M2 kept as sensitivity checks. Testing also exposed a worker-direction coding error (slide 24).

SYNTHETIC DATA · tests the method, not Xinhuang ridership

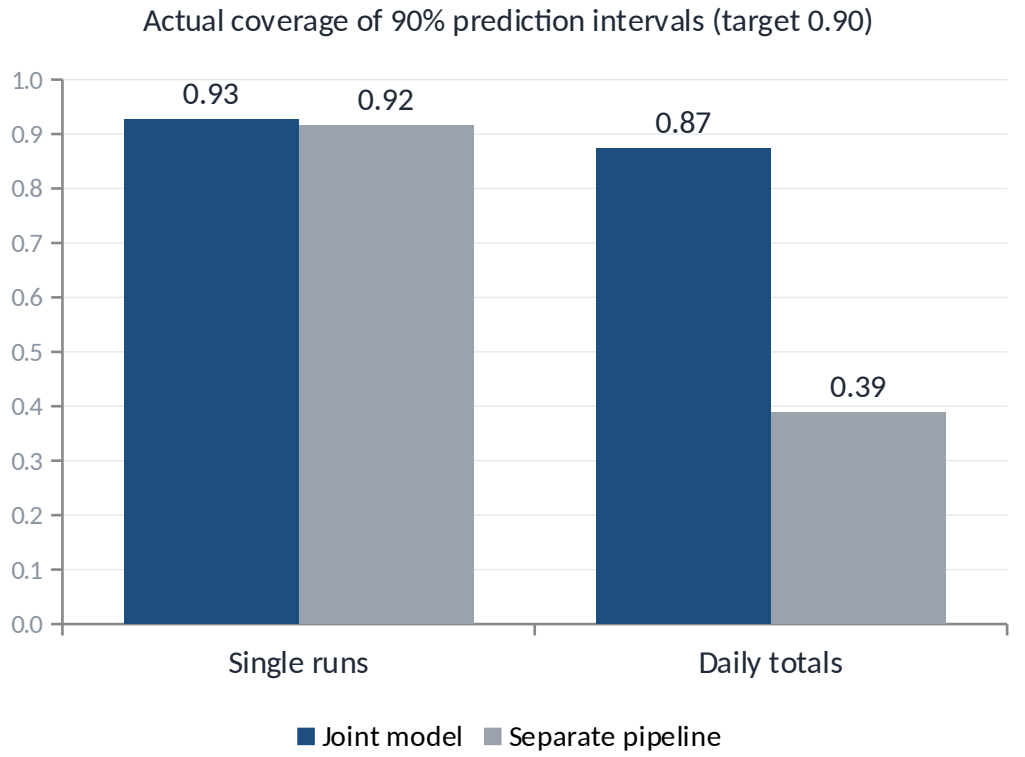
Forecasting every run 1–7 days ahead — and keeping daily totals honest

HOW A FORECAST IS DRAWN (REPEATED 300 TIMES)

1. Draw model parameters
2. Draw one county-wide shock per forecast day
3. Draw turnover κ per route
4. Draw each run's latent demand; cap at capacity
5. Sum runs within the same draw → daily total

EVALUATION

- Rolling origins on exploration data (Jun 2025, Dec 2025, Jun 2026); CRPS, 90% coverage overall and by route, time of day and event; daily-total MAPE.
- Baselines: seasonal naïve (same weekday and run, past four weeks) and the separate-stage pipeline.
- One sealed test: the Oct 2026–Feb 2027 ledger (incl. Spring Festival 2027) is opened once, after freezing.



World A, 8 replicates. Daily intervals average 566 passengers (joint) vs 194 (pipeline): the pipeline's narrower interval is wrong, because runs on the same day share shocks. Run-level CRPS is almost identical (2.82 vs 2.84).

SYNTHETIC DATA · tests the method, not Xinhuang ridership

From forecasts to a fixed-core, flexible-supplement service

THREE SERVICE LAYERS (TOP LAYERS VARY DAY BY DAY)

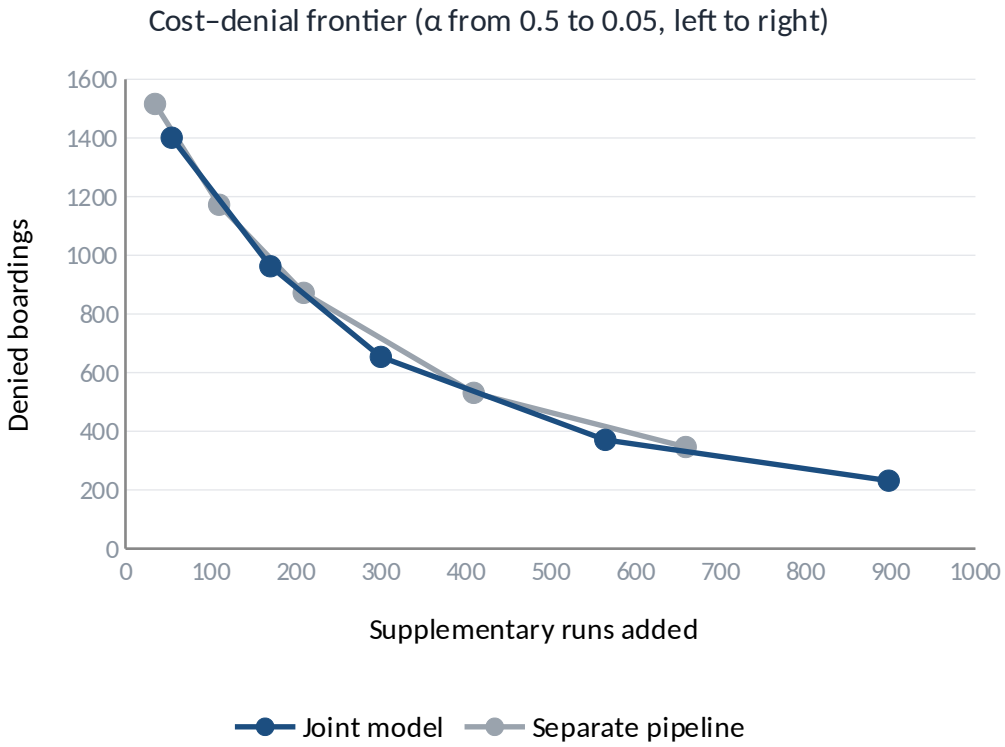
Forecast-triggered runs
Add a run next to scheduled run i if Pr(demand > κ × seats) > α — a chance constraint solved on forecast samples.

Day-ahead reserve
Buses and drivers held for tomorrow, sized from an upper quantile (e.g., 90%) of the daily-total forecast beyond core capacity.

Fixed core timetable
Stable, no booking; guarantees basic service on every route — essential for older riders.

SURVEY INPUTS TO THE PLAN

Preferred departure times (schedule delay) · what denied riders do — wait, switch mode, give up — which weights the cost of denial · how riders learn of extra runs (village WeChat groups, notices, drivers).



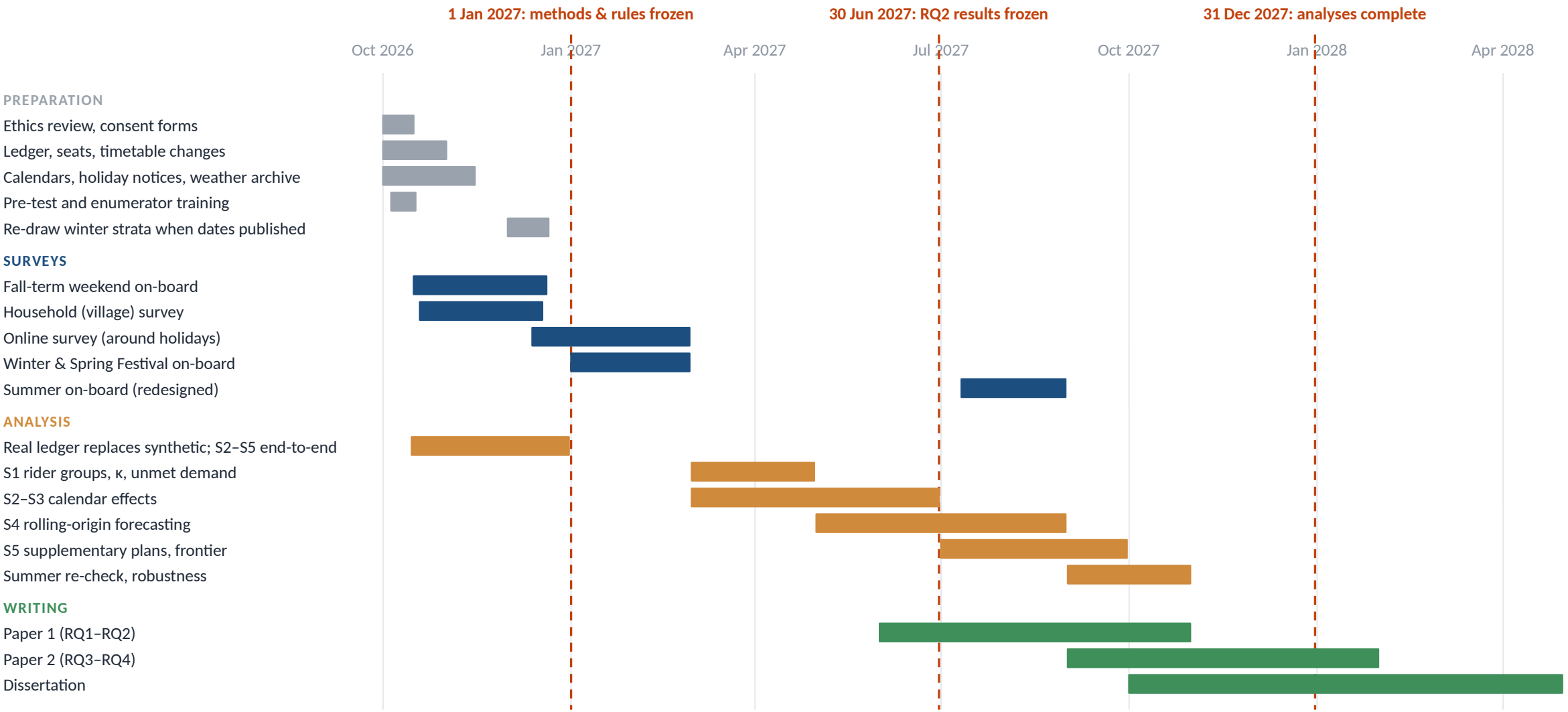
Both approaches lie on nearly the same frontier; at equal α the joint model adds more runs and catches more overloads (α = 0.05: 85% vs 76%). Its real advantage is a trustworthy α and a calibrated reserve. The county chooses α from the cost of an extra run vs. the social cost of a denied rider. Plans are compared with the current timetable and a simple calendar rule.

Hypotheses and how results will be read — written before the data

Hypothesis	If supported	If not supported
H1 (RQ1) Boarding students and weekly workers dominate in-term weekend runs; far villages give up more trips	Design core runs and triggers around these groups; prioritise far villages	Revise rider groups and encodings; unmet demand has other causes
H2 (RQ2) All five factors have positive independent effects, separable at run level	Group-, direction- and time-specific supplementary rules	Group small or shifted to other modes — explained with surveys
H3 (RQ2) S×R and CY×T overlaps are negative (the same people travel once)	Do not add factor effects on overlap days	Positive: extra capacity needed; ≈0: add effects
H4 (RQ2) Holiday returns grow with holiday length; type effects follow prior importance	Tier holiday supplements by type × length	Use data-estimated type effects
H5 (RQ3) Run and daily 90% coverage ≈0.90; CRPS beats seasonal naïve	Forecasts usable 1–7 days ahead for both layers	Add route-level shocks; else calendar rules only
H6 (RQ4) At equal extra runs, the two-layer plan denies fewer riders than the current timetable	Recommend fixed core + reserve + triggered runs	Recommend the simple calendar rule; report limited value

Rules: report null and non-significant results with intervals; distinguish "no effect" (narrow interval around zero) from "not identifiable" (wide interval, low support); if main and sensitivity methods disagree, report both.

Timeline: October 2026 – April 2028



Immediate next steps (Oct–Dec 2026) and main risks

CHECKLIST

1. Confirm ledger fields with the county bus station; obtain exploration-period ledger and per-run seat counts; add a “full bus, riders left behind” tick box
2. Submit ethics review; prepare consent script and enumerator manual
3. Pre-test each instrument on 1–2 routes (20–30 each): comprehension, timing, skip logic
4. Run the fall-term weekend on-board wave and the village household survey
5. Swap synthetic for real ledger; run S2–S5 end-to-end; confirm the frozen method
6. Re-draw winter strata once 2027 holiday dates are published
7. Freeze methods and interpretation rules by 31 Dec 2026 with a time-stamped registration

Risk	Consequence	Mitigation
Ledger only as daily totals	RQ2–RQ4 cannot be answered per run	Secure first; else daily model, run level as scenarios
2027 winter dates announced late	Winter strata drawn on assumed dates	Re-draw with the same seed; non-winter strata unchanged
Too few enumerators in the Spring Festival peak	Peak strata under-sampled	Peak strata first; adjust weights
Poor tally-sheet quality	κ wrong → censoring and decisions wrong	Daily balance checks; κ at 10th/90th percentiles
Sealed test disappoints	H5 not supported	Report by pre-set rules; no re-testing on the same data
Costs and spare buses unverifiable	S5 plans not implementable	Frontier under two cost scenarios

Expected contributions and outputs

Substantive

- First run-level evidence on how rural calendar rhythms — boarding-student and worker weekly trips, term surges, holiday returns, Spring Festival — shape county bus demand.
- Where and for whom the fixed timetable leaves demand unmet, by village accessibility.

Methodological

- Identifiability-screened estimation of independent vs. combined effects on the passenger scale, with capacity censoring.
- A joint model whose run forecasts add up to calibrated daily totals.
- Estimator chosen by synthetic backtest and frozen before testing.

Practical

- A fixed-core + reserve + forecast-triggered rule with a cost–denial frontier for the county transport bureau.
- Reproducible code and data pipeline that other counties can adapt.

PLANNED OUTPUTS

- **Paper 1 (RQ1–RQ2):** independent and overlap effects of calendar rhythms on rural bus runs — Journal of Transport Geography / Travel Behaviour and Society / Transportation Research Part A.
- **Paper 2 (RQ3–RQ4):** run-level probabilistic forecasting and chance-constrained supplementary runs — Public Transport / Transportation Research Record / TR Part A.
- **Dissertation:** nine chapters, including the full method comparison and failures.

Limitations — and questions I would value your input on

KNOWN LIMITATIONS

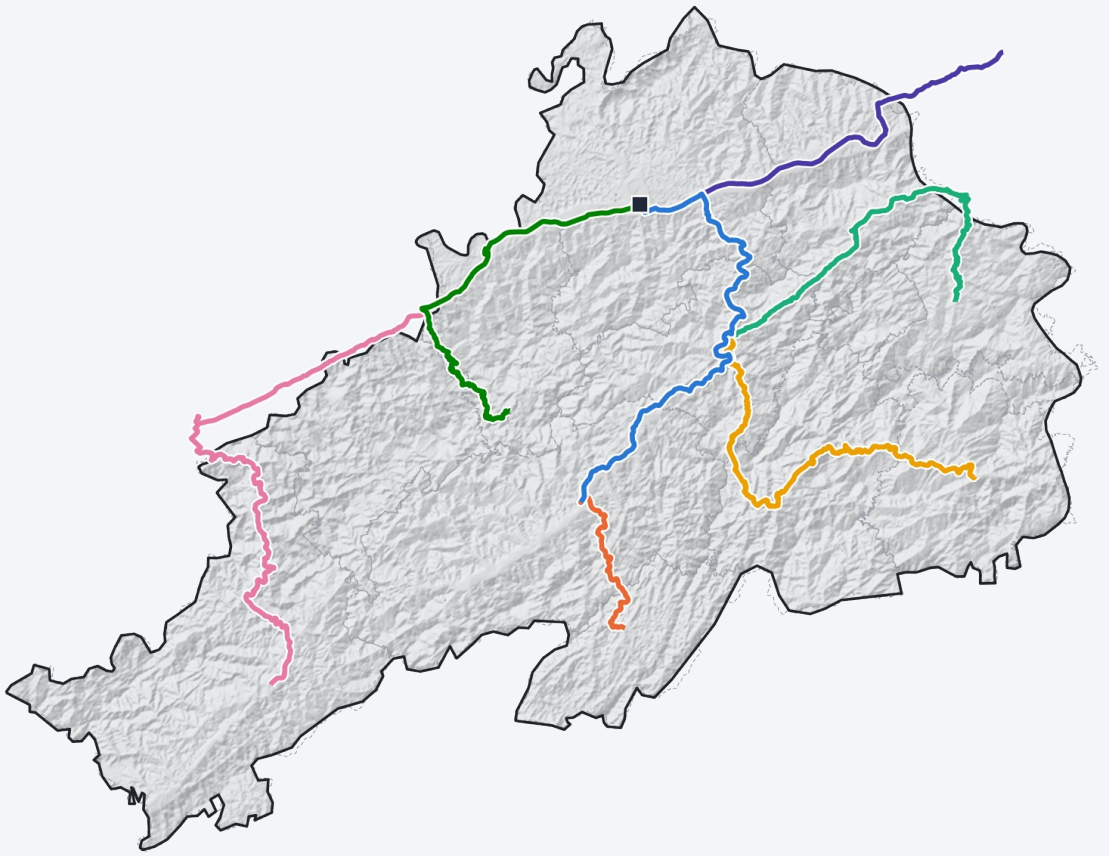
- Synthetic results show that the method recovers known truth; they are not findings about Xinhuang.
- Three years of calendar give limited support for some overlaps (W×R: 69 runs); only a longer ledger fixes this.
- Turnover κ from sampled tally sheets propagates into censoring and decisions.
- Unobtainable drivers (industrial shifts, clinic rosters) may leak into weekday or residual terms.
- The sealed test contains one winter and one Spring Festival.
- Chinese literature: the systematic CNKI search is still to be run.

QUESTIONS FOR THE AUDIENCE

1. Is the passenger (additive) scale the right primary estimand for operators, with the ratio scale as secondary?
2. Is ≥200 runs a sensible identifiability threshold — or should support be judged by interval width?
3. Beyond forecast-triggered extra runs, which supplementary service types (route deviation to far villages, scheduled on-call runs) merit testing in a Chinese mountain county?
4. How can latent demand be validated when denied boardings are only partly recorded?

Thank you

- Run-level, direction-aware calendar factors separate rider groups that daily totals conflate.
- Overlapping effects are estimated only where the calendar can identify them, on the passenger scale, with censoring.
- One joint model turns effects into calibrated forecasts and a transparent cost–denial choice for supplementary service.



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A1 · Questionnaire indicator groups and the questions they serve

Group	Indicator group	Items	Serves
G0	Run key and non-response record	Z1–Z4, H0	RQ1, RQ2
G1	Trip origin–destination and trip chain	A1, A2, A5, H7, O8	RQ1, RQ4
G2	Rider type and periodic rhythm (maps to S, W, T, R/CY)	A3	RQ2
G3	Trip purpose	A4	RQ1
G4	Time constraints and preferred departure	A6, H11, O10	RQ4
G5	Dependence and alternatives	A8, H8	RQ4
G6	Unmet demand	A9, H6, H9, H10, O9	RQ1, RQ4
G7	Service-gap priority (forced choice)	A10, H12, O12	RQ1
G8	Demographics and weights	A7, A11–A13, H14, O2, O3	RQ1, RQ2
G9	Household structure and group exposure	H1–H5	RQ2
G10	Holiday return timing	S15, W15, O4–O7	RQ2, RQ3
G11	Acceptance of extra runs; information channels	A14, H13, O11	RQ4
G12	Channel and duplicate responses	O1	RQ1

- Prefixes: A on-board, H household, O online, S/W summer/winter on-board, Z enumerator record.
- Not included, with reasons: theory-of-planned-behaviour scales (no RQ needs intentions; behaviour is observed directly), SERVPERF batteries (burden), shift and clinic rosters (unobtainable), waiting time (reflects departure choice on hail-and-ride routes).
- Reliability and validity without multi-item scales: test–retest $\kappa \geq 0.6$ on A3, A4, A7; inter-coder κ on field-coded items; tally-sheet totals vs. ledger; pre-registered construct checks.

A2 · On-board sampling strata (runs, days and planned questionnaires)

Wave	Stratum	Runs in stratum	Days	Sampled runs	Planned questionnaires
Fall 2026	**A1 Last school day p.m. · outbound (students home)**	330	10	5	40
Fall 2026	**A2 Last workday p.m. · inbound (workers to town)**	320	10	5	40
Fall 2026	**A3 Eve of school week p.m. · inbound (students back)**	320	10	5	40
Fall 2026	**A4 Eve of work week p.m. · outbound (workers out)**	330	10	5	40
Fall 2026	**A5 First rest day a.m. · inbound (workers to town)**	360	10	5	40
Fall 2026	**A6 Tue–Thu p.m. · both directions (control)**	1,755	27	5	40
Winter 2027	**B1 Spring Festival pre-peak · outbound (going home)**	680	10	39	312
Winter 2027	**B2 Spring Festival post-peak · inbound (returning)**	816	12	39	312
Winter 2027	**B3 New Year holiday first/last days · both**	204	3	13	104
Winter 2027	**B4 Weekly-commute windows outside Spring Festival · both**	675	18	23	184
Winter 2027	**B5 Tue–Thu p.m. · both directions (control)**	943	23	15	120

Equal allocation across fall strata (each is a separate check); winter allocation follows research priority (Spring Festival peaks 30% each). About 8 questionnaires per run is an assumption to be checked in the pre-test. 98 of 129 winter runs fall on dates marked unverified and will be re-drawn.

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